Vinay Dinkar Kale

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Senior Data Scientist with 8+ years of experience specializing in advanced ML modeling, time-series forecasting, and deploying production-grade Generative AI and Agentic solutions.

Experience

- **Amazon (AWS)** — *Santa Clara, USA*
  Senior Data Scientist — *July 2021 - Present*
  - Defined the strategic science roadmap for AWS Connect's AI initiatives, aligning high-impact machine learning projects with core enterprise business objectives.
  - Spearheaded the R&D of GenAI capabilities and LLM-based agents to automate complex supervisor workflows, serving as the technical anchor for global predictive systems.
  - Architected and scaled novel AI forecasting and anomaly detection models for core demand/supply metrics, optimizing agent allocation for 300+ government agencies and Fortune 100 companies.
  - Authored patented ML methodologies for sparse data time-series forecasting, while coaching peer scientists to drive organizational excellence in model architecture and validation standards.
- **Capital One** — *NYC, USA*
  Senior Data Scientist — *Feb 2019 - July 2021*
  - Developed robust and reusable next-generation valuation, risk, and fraud XGBoost ML model pipelines reaching 100M+ customers, with an estimated incremental value of $35M per year.
  - Architected an agile and automated monitoring framework for valuation models, decreasing the time required to complete quarterly compliance monitoring from one week to 8 hours.
  - Built Python libraries for data sourcing and cleaning with a focus on design, reusability, and user experience. Conducted multiple refactor cycles to maintain efficiency and keep systems up-to-date.
- **Spotify** — *NYC, USA*
  Data Science Intern — *Jun 2018 - Aug 2018*
  - Developed a fan-artist pair segmentation pipeline that quantifies the affinity for 20 billion fan-artist pairs and analyzed how a new feature affects fan-artist journeys through statistical analysis on data from A/B tests.
- **ZS Associates** — *Pune, India*
  Data Scientist — *June 2016 - July 2017*
  - Developed a marketing-mix modeling pipeline using multivariate regression for a client in the retail media sector. Increased total ROI by 15 percent and secured engagements for 3 additional similar projects.
  - Built a novel patient clustering engine for pharmaceutical clients using patient-data vector embeddings trained on sequential big medical data. This served as input to several patient-level classification models.

Awards and Research

Patents

- US-12511576-B1: Correcting Time Series Predictions for Seasonality with Trends in Contact Centers (Granted)
- P86851-US01: Accurate Average Speed of Answer (ASA) and Service Level (SL) Forecasts using Effective Supply calculation in Contact Centers (Filed)

Publications

- Computer Vision and Pattern Recognition (CVPR) Conference 2018 : Traffic Surveillance Research Paper. We introduce novel usage of Mask-RCNN algorithm for vehicle detection, speed estimation and vehicle tracking for video footage of cars in highway. Achieved 100 percent detection rate and 7.97mi/hr speed estimation error Citation Count: 15.
- Amazon Machine Learning Conference (AMLC) 2023: Generalized Anomaly Detection with Contact Center Use Case: A research paper reviewing various anomaly detection techniques and proposing a novel ensemble approach tailored for identifying anomalies in contact center environments.

Awards

- National Interest Waiver (NIW): Granted EB2-NIW by the US government to pursue the endeavor of "Improving efficiency of contact centers by using Artificial Intelligence (AI) driven Work-Force Management (WFM) systems" in June 2025.
- Runner Up, Columbia Data Science Society Hackathon: Built natural language processing network model to find fraud in Enron email data corpus. Won cash prize of 2,000 dollars. September 2017

Education

- **Columbia University** *NYC, New York*
  M.S. Data Science *Sep 2017 - Dec 2018*
  Relevant Coursework: Probability Theory, Statistics, Machine Learning, NLP, Deep Learning, Algorithms, Data Storytelling
- **Indian Institute of Technology Madras (IITM)** *Chennai, India*
  B.Tech Mechanical Engineering, M.Tech Product Design *Aug 2011 - May 2016*
  Relevant Coursework: Computational Engineering, Time Series Analysis, Regression Models, Operations Managment

Skills

- **Languages** Python, R, SQL, Spark, C, C++, LaTeX
- **Frameworks** Scikit, Dask, TensorFlow, PyTorch, Tableau, MS Office, LLMs, Agentic AI.
- **Platforms** Docker, GIT, Airflow, MySQL, Linux, AWS, Microsoft Azure, Google Cloud

Volunteer Experience

- **Organizing Committee, PyData Conference NYC** *June 2019 - Sep 2019*
  Led team for reviewing proposals and spearheaded the Diversity Scholarship Initiative
- **Teaching Assistant, Columbia University** *Jan-Apr/Aug-Dec 2018*
  COMS 4995 Applied Machine Learning, COMS 4996 Applied Deep Learning.